Pediatrics

Workbook

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
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DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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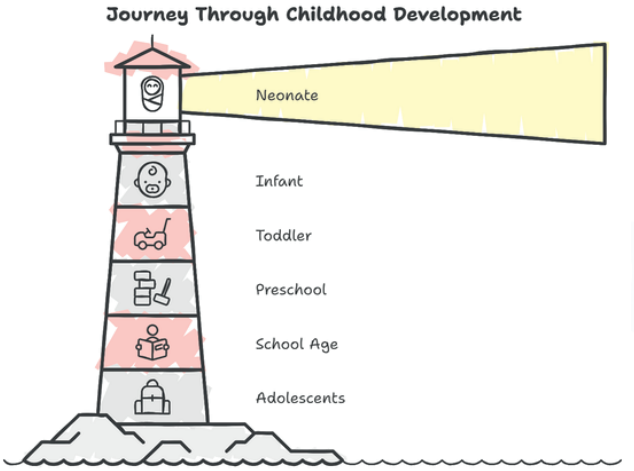
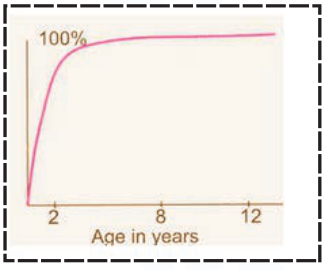
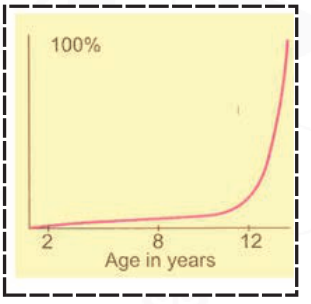
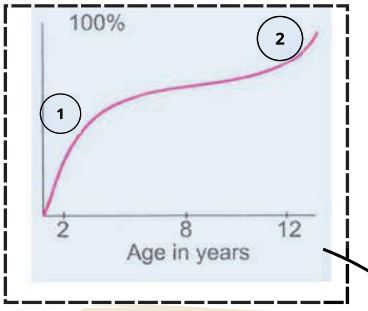
Chapter 1 : Growth and Development



Growth patterns







Length/ Height

Stadiometer

Infantometer

Value at birth:

Double

Triple

PUBERTY

Male	Female

Which of the following closely corresponds with sexual maturation?

A) Sexual maturation
B) Brain growth
C) Lymphoid growth
D) School age

During puberty, boys gain about 20-30 cm and girls about 16-28 cm.

Final conculsions....

- Puberty occurs early in _____
- Pubertal growth spurt occurs early in _____
- Final (Adult) height is more in _____

Weight

Birth weight (average) –

Double –

Triple –

Quadruple –

Weight loss after birth?

Reason:

Identify:

Age group?

Usefulness?

Value of MAC:

Head circumference

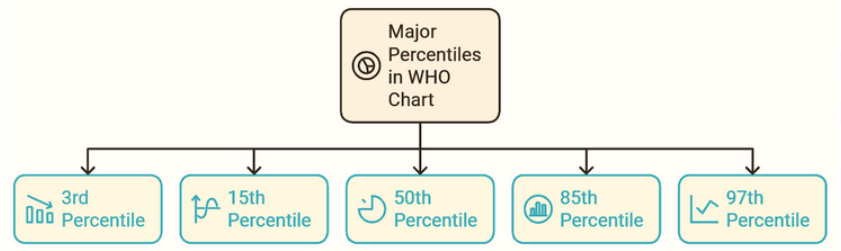
WHO CHARTS

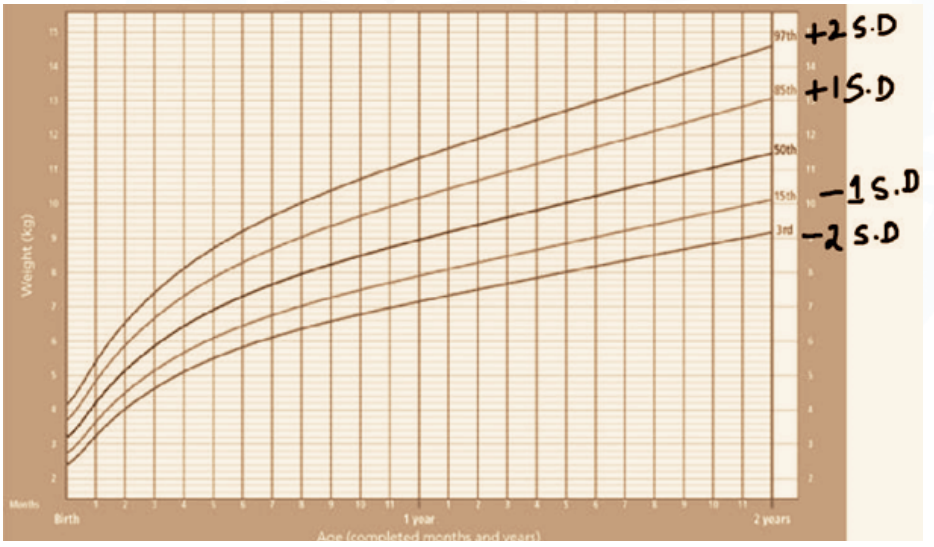
Variants

PERCENTILES S.D

PERCENTILES

S.D





Interpretation:

	Percentile	S.D
Low		
High		

Dentition

	Primary teeth	Permanent teeth
Number		
First teeth to erupt		
Time of eruption		

Period of mixed dentition?

SHORT STATURE

Definition:

Types

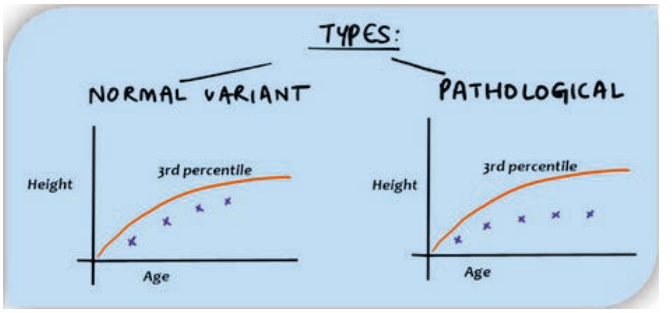
- Normal variants
- Pathological

Pathological short stature

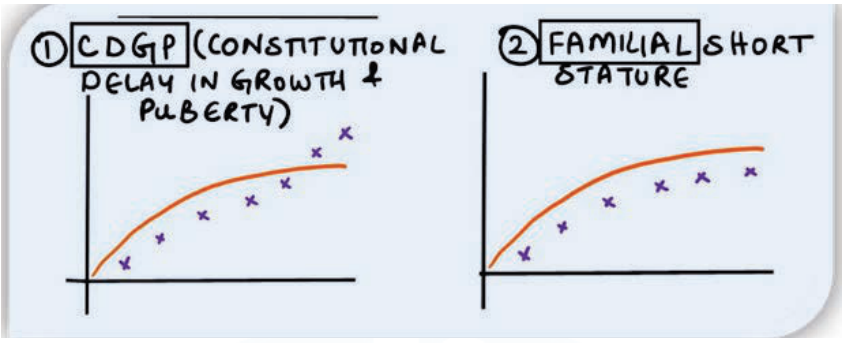
1.
2.
3.
4.

Expected height in short child:

MPH (Mid parental height):



Normal Variants



	Constitutional delay	Familial short stature
Final (Adult) height		
Puberty		
Bone Age		
Parents' height		

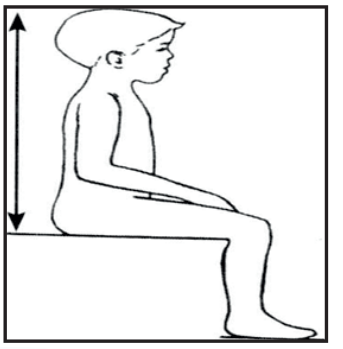
How to determine the bone age?

Overall m/c cause of short stature (and delayed puberty)?

Delayed puberty: Definition

Age	X ray
Newborn	Knee
Infant	Shoulder
1 - 13 years	Hand & wrist

Identify:



Normal values:

At birth

3 years

7 years

>7 years

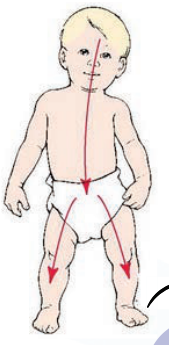
Short Stature: US: LS Ratio

Normal Ratio

Abnormal Ratio

DEVELOPMENTAL MILESTONES

Gross motor milestones



1. Gross Motor
2. Fine Motor
3. Language
4. Social

Rule:

3 months

4 months

6 months

8 months

10 months

12 months

15 months

18 months

2 years

3 years

4 years

Fine motor milestones

Rule:

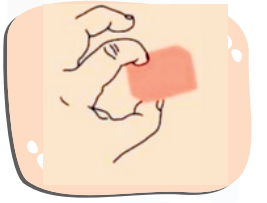
3 months

4 months

6 months

8 months

10 months



12 months

15 months

18 months

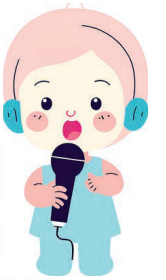
2 years

Copying shapes

Dressing skills

Handedness

Language milestones



3 Months

6 months

9 months

12 months

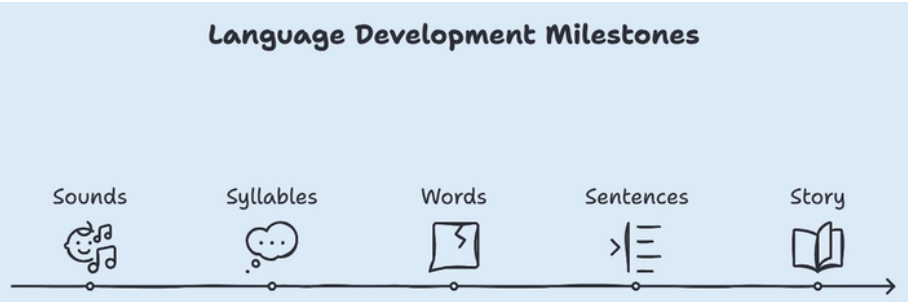
15 months

18 months

2 years

3 years

4 years



Social milestones

2 Months

3 months

4 months

6 months

9 months

12 months

3 years

4 years

Identify this condition in 6 year old boy

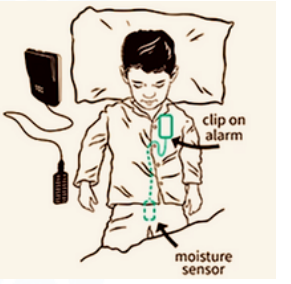
Definition:

Likely reason?

Management

Initial:

- Voiding before going to bed
- Fluid restriction (evening)
- Avoid caffeinated drinks



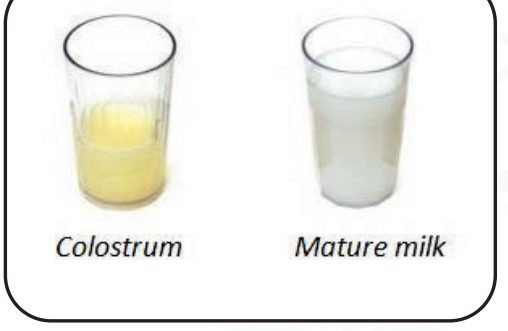
Drug treatment:

Chapter 2: Nutrition

Breastfeeding:

When to start?

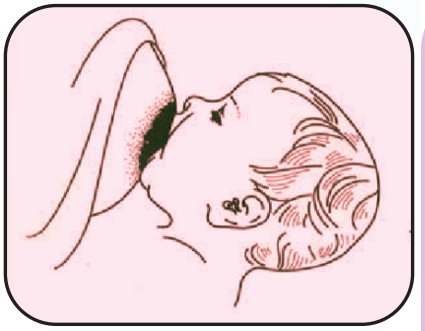
Colostrum → Transitional Milk → Mature Milk



Properties of colostrum

Colour:

3 'A's



Signs of good attachment (Latching)

1.

2.

3. (Upper areola may be visible)

Rooting reflex

Appears by 32 weeks
Fully developed by 34 weeks

Modes of feeding in preterm babies:

Gestational age	Mode of nutrition
< 32 weeks	
32 – 34 weeks	
> 34 weeks	

Properties:

Nutritive value:_____kcal/ 100mL

Carbohydrate:

- More in breast milk
- Major carbohydrate:

Protein:

- ___ times less in breast milk

Predominant protein:

(Cow milk –

Important whey proteins are Lactoalbumin and lactoglobulin

Aminoacids in breast milk:

Fats:
~ almost same as in cow milk

Breast milk is rich in PUFA

Vitamin Deficiencies in breast milk:

Contraindications for Breast feeding (Absolute)

Baby	Mother

! Most maternal infections are relative contraindications

Expressed breast milk Storage:

Room Temperature

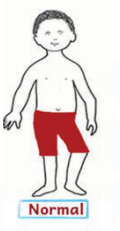
Refrigerator

Freezer (-20 C)

Immunity from breast milk:

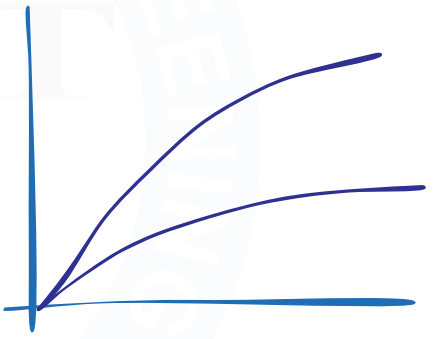
1. IgA
2. Lactoferrin
3. Lysozyme
4. Bifidus factor

MALNUTRITION: Basic concepts



Severity of Malnutrition:

Moderate malnutrition	Severe malnutrition



Important conditions under Severe Malnutrition:

SAM (Severe Acute Malnutrition)

1.

2.

3.

Presence of any one of the features is sufficient

Complicated SAM vs Uncomplicated SAM

Major complications in a child with SAM:

1. Sugars
2.
3. Electrolyte imbalances

Cut off value:

_____ (Neonate

? Fluid of choice in a child with shock and SAM?

Kwashiorkor ('Fat sugar' baby)

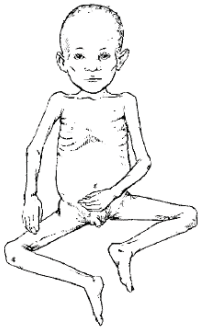
Predominant deficiency:

Hallmark:

Appearance:

Appetite:

Marasmus (= Adaptive starvation)



Predominant deficiency:

Hallmark:

Appearance:

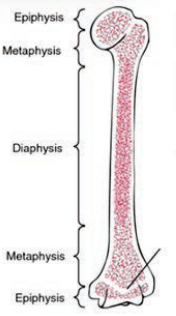
Appetite:

Additional findings (Kwashiorkor)

Micronutrient deficiency disorders

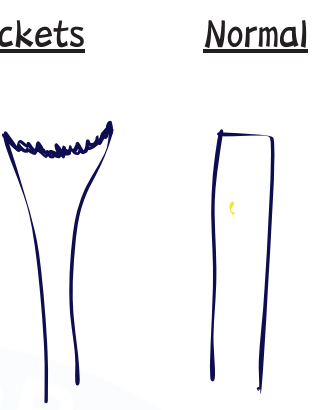
Rickets:

Site of defect?



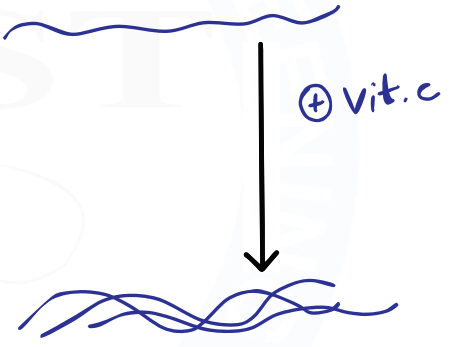
Basic defect

Features:

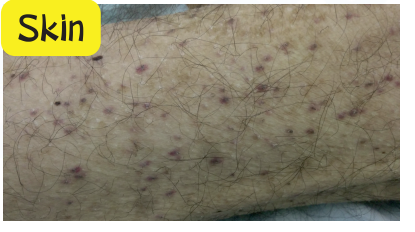


Scurvy:

Basic defect



Features:

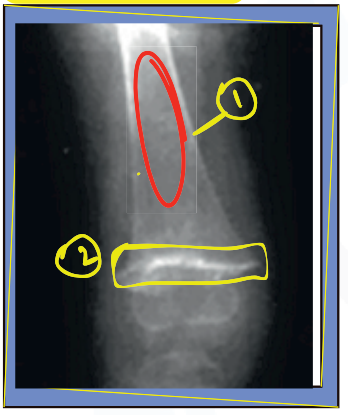


Rosary: Rickets vs Scurvy

Rickets

Scurvy

X ray: Scurvy



Disorders with Rash + Diarrhea

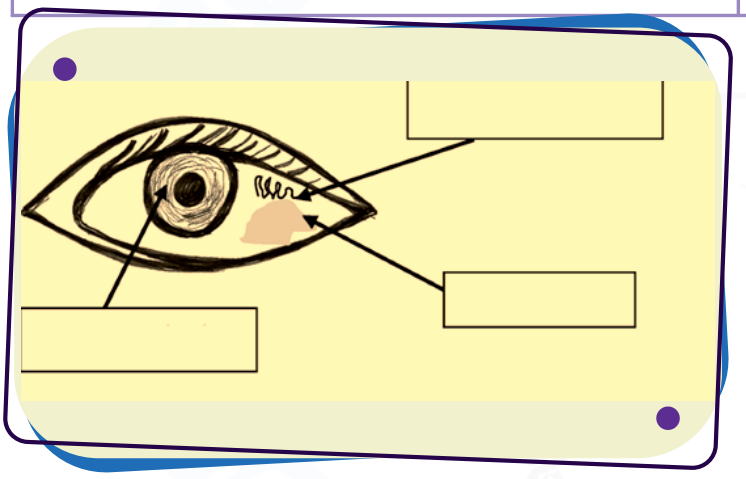
Diagnosis

Diagnosis

Defect:

Identify the most probable micronutrient deficiency in each of the situations given below:

Poor wound healing	
Easily breakable hairs, low IQ and ATP 7A gene defect	
Difficulty in dark vision, dry eyes (xerophthalmia) and dry skin	



Vitamin A dose (IU):

< 6 months

6 – 12 months

> 12 months

(Weight < 8 Kg

Treatment Schedule (in Malnutrition)

Chapter 3: Genetic Disorders

Chromosomal Aneuploidies:

1.

2.

3.

Trisomies

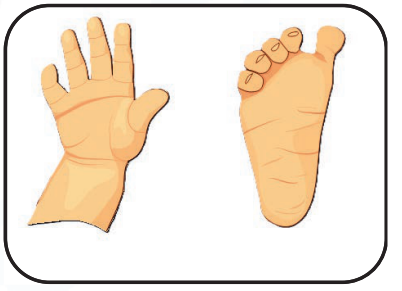
Overall m/c

2nd m/c trisomy

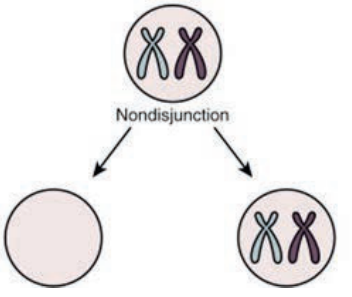
m/c trisomy in spontaneous abortion

While evaluating a child with short stature and intellectual disability, the following finding is noted on examination. What is the likely diagnosis?

a) Trisomy 21
b) Trisomy 18
c) Edward syndrome
d) Patau syndrome



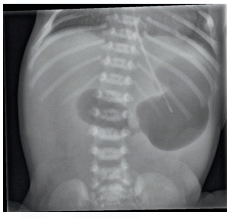
Genetic defect



Down syndrome facies

Organ defects:

- m/c –
- Brain – Low IQ, Microcephaly
- GIT –
- m/c leukemia –



Antenatal Diagnosis

"Best overall: "

others : USG, Markers

Remember as 'High'

H –

I –

Other trisomies: Short overview

Trisomy 18

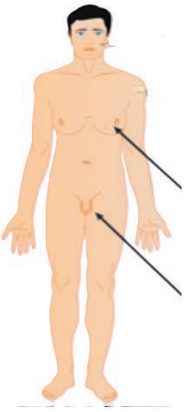
Trisomy 13

?

You are evaluating a tall male in view of delayed puberty. He is also noted to have intellectual disability and learning difficulties. Karyotyping reports are given below. Which of the following is not the likely abnormality to be noted in this boy?

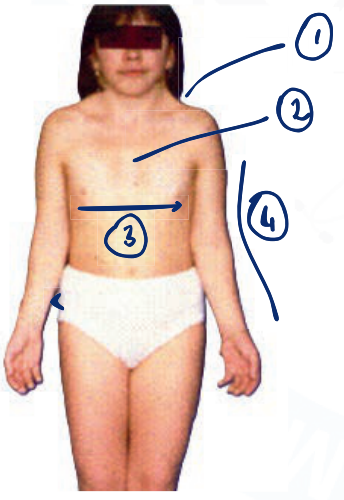
a) Low testosterone
b) High LH and FSH
c) Macroorchidism
d) Gynaecomastia

Diagnosis:



This condition is the most common cause of sex chromosomal aneuploidy in males

Turner syndrome



Findings:

1. _____
2. _____
3. _____
4. _____

Gonads: _____

IQ -

Which condition resembles physical appearance of Turner syndrome but affects <u>both males and females</u>?

IQ -

Gonads (Male):

Most common inherited genetic disorder with Intellectual disability?

Chapter 4 - Neonatology

Essential Newborn care after birth:

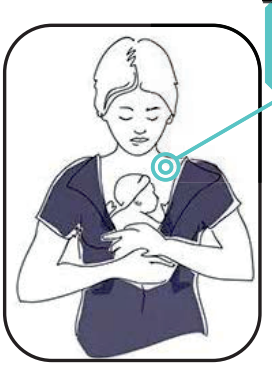
Which of the following is not included in newborn care after birth?
a) Cleaning the baby
b) Breastfeeding
c) Ear care
d) Delayed cord clamping

Other components of newborn care:
- Immediate Skin-to-Skin Contact
- Vitamin K administration
- Temperature Maintenance

Normal temperature:

Site:

Identify this intervention to protect the newborn from hypothermia



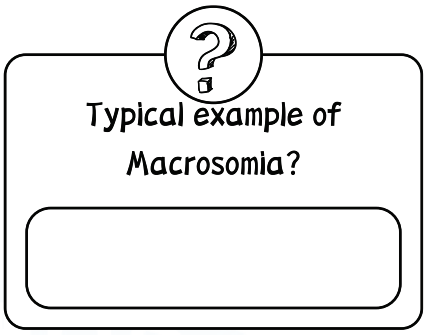
DON'T FORGET!
Head should be separately covered → Importance:

Indications

Posture: _____

Some numbers to remember:

- LBW –
- VLBW –
- ELBW –
- Macrosomia –
- (Big baby)



Which of the following complications is common in infant of diabetic mother (IDM)?

a) Anemia
b) Hypoglycemia
c) Hyperglycemia
d) Macrocephaly

Infant of diabetic mother (IDM)

IUGR = Intrauterine growth restriction

- Small baby with evidence of wasting
- Assessment: PONDERAL index (PI)

Role of APGAR score:

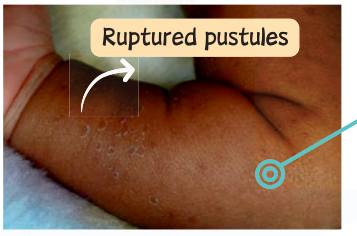
SCORE	APPEARANCE	PULSE	GRIMACE	ACTIVITY	RESPIRATION
0	Blue all over	No pulse	No response to stimulation	No movement	No respiration
1	Blue extremities	<100 beats/min	Grimace on stimulation	Some flexion	Weak, irregular, slow
2	No blue colouration	>100 beats/min	Cry on stimulation	Flexed limbs that resist extension	Strong cry

Normal scores: (out of 10)

! Low APGAR scores are noted in birth asphyxia

Normal observations in newborn

- Most common rash in newborn
- Appears > 24 hrs after birth



(Appears within 24 hrs)

Identify this finding noted in the palate of a normal newborn?

You are evaluating a 12 hour old newborn with the following finding. What should be the management?

Reason:

HEAD SWELLINGS:

	Caput succedaneum	Cephalhematoma
Location		
Reason		
Content		
Time of appearance		

- Normal time for passage of:
  - Urine
  - Meconium

NEONATAL REFLEXES: (also called as primitive/ immature reflexes)

Which among the following is the earliest to appear?
a) Rooting reflex
b) Palmar grasp reflex
c) Parachute reflex
d) Transfer objects from one hand to another

Identify:

Components:

Duration:

NEONATAL RESUSCITATION

Basics

NO CRY at birth ⟶ start Initial steps:

Correct position:

After initial steps, if no response – assess heart rate

Why?

Normal heart rate?

Next step:

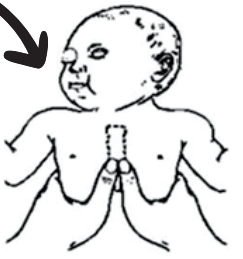
Non invasive

Invasive

Initial O2 concentration during PPV?

First response to PPV?

No response to PPV ⟶ Next step:



Summary of Neonatal Resuscitation

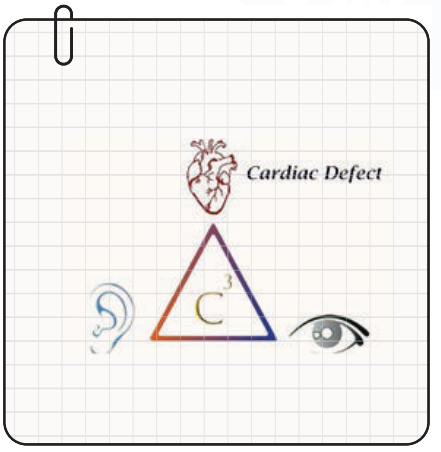
Initial assessment
↓
Initial action (steps) → T A B
↓
Assess HR and Respiratory effort
↓
PPV
↓
Chest compression (with ventilation)
↓
Adrenaline

NEONATAL DISORDERS

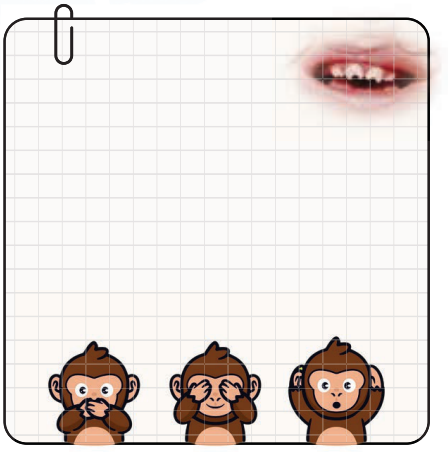
INTRAUTERINE INFECTIONS

- Toxoplasmosis
- Others →
- Rubella
- CMV
- Herpes

Congenital Rubella ~ 3 'C's



Hutchinson's triad



While evaluating a term newborn you notice continuous nasal discharge, abdominal distension and palmoplantar blisters. Which test will establish diagnosis in this case?

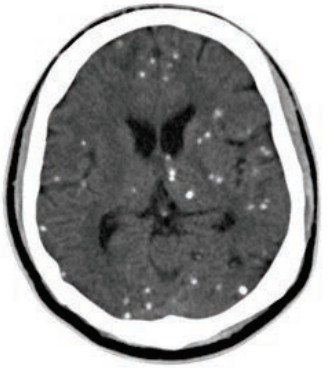
Features:

Diagnosis?

Other characteristic finding is intense bone pain (limbs) →

Radiological associations of TORCH infections

Brain imaging



Diagnosis:

Diagnosis:

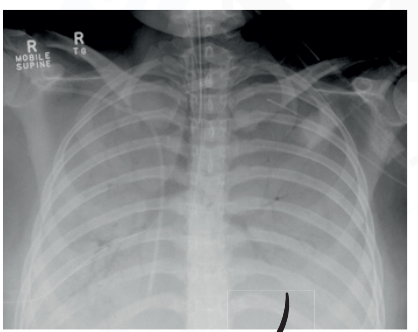
One liners:

- Most common cause of neonatal sepsis?
- Mother complaints of a 3 day old neonate having crying while passing urine. Likely possibility?
  (m/c cause:

Identify the cause of RESPIRATORY DISTRESS

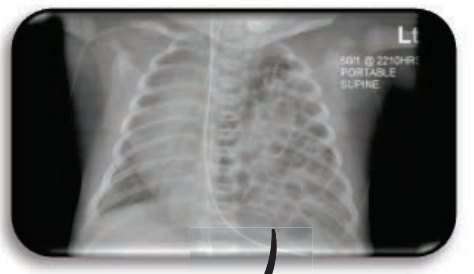
32 week preterm baby
Diagnosis:

Pathophysiology



Findings:

Antenatally detected anomaly with following X ray



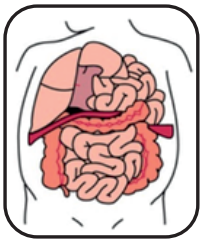
Findings:

Pathophysiology

Why?

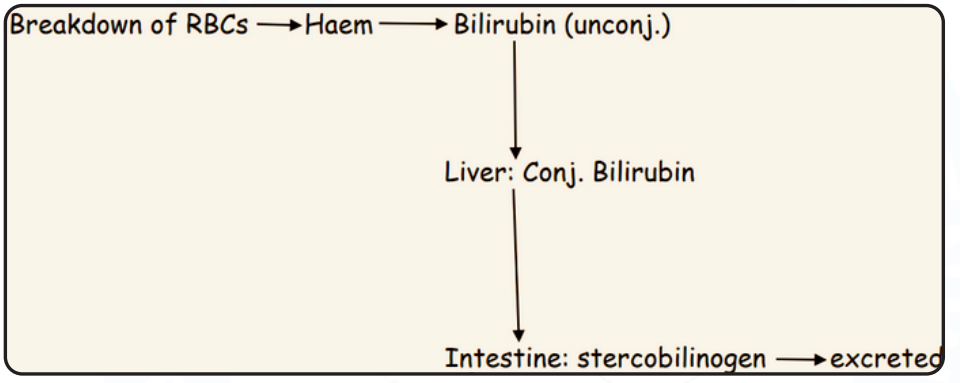
Which step of neonatal resuscitation is C/I in CDH babies?

Which of the following is not a feature of CDH?
a) Respiratory distress
b) More common on left side
c) Sunken abdomen
d) Dextrocardia



NEONATAL JAUNDICE

Bilirubin metabolism



Assessment of jaundice

- Best
- Screening

Normal values

Total	Direct	Indirect

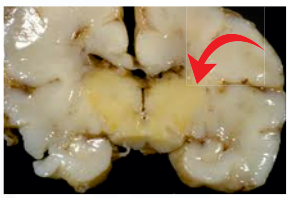
Rule:

Area	Approx. level of bilirubin (mg/dL)
1st	
Last	

DANGER

Bilirubin encephalopathy can lead to long term sequelae of cerebral palsy → Type?

Target area:



Neonatal jaundice

- Physiological
- Pathological

Physiological jaundice

Why normal jaundice?

- More RBCs
- Less life span of RBCs
- Immaturity of UDP – GT

Criteria for physiological jaundice

1. Appearance
2. Duration
3. Area NOT affected

If none of these are noted, then it is PATHOLOGICAL jaundice

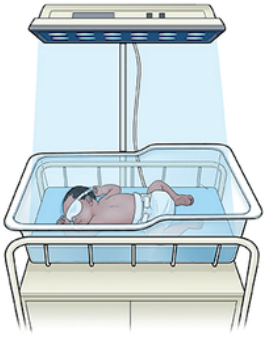
Pathological jaundice: Etiology (↑ Unconjugated bilirubin)

↑ Production	↓ Conjugation

Also noted in GILBERT syndrome but it is of late onset (adolescents)

Treatment

Phototherapy



Recent exam topic:

Fluid requirements in neonate (mL/Kg/day)

Birth weight	Day 1	Day 2	Day 3	Day 4	Day 5
≥ 1500 g	60	75	90	105	120

What is the amount of milk given to a term baby in first 24 hours after delivery?

a. 40 mL/kg

b. 50 mL/kg

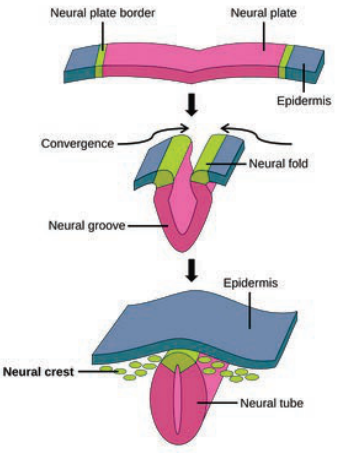
c. 60 mL/kg

d. 80 mL/kg

Chapter 5: Neurology

Neural Tube Defects (NTD)

Basic defect:



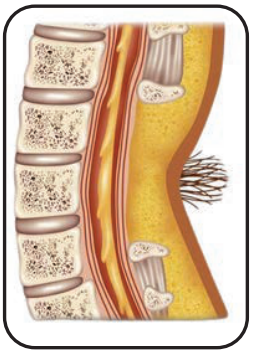
Time of gestation?

_____

Caudal NTDs

Usual location

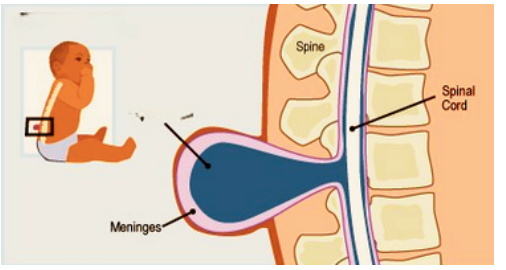
1



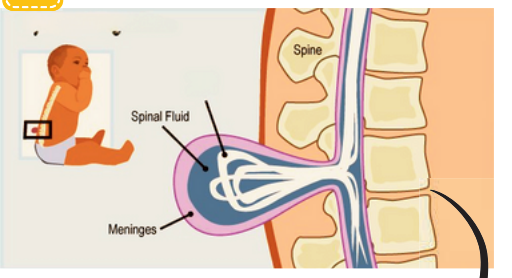
Clues (overlying defect):

This is the most common and most mild form of NTD

2



3



Additional feature:

Common identifying features:

- Swelling (mass) in the back ~ midline
-

Cranial NTD:

- Identify:

Risk factors (Multifactorial Inheritance)	Antenatal diagnosis

Prevention aspects

Dose:

(if mother has previously affected child -

Seizure disorders in children

	Febrile seizures	Absence seizures
Age		
Description		
EEG		
Treatment		

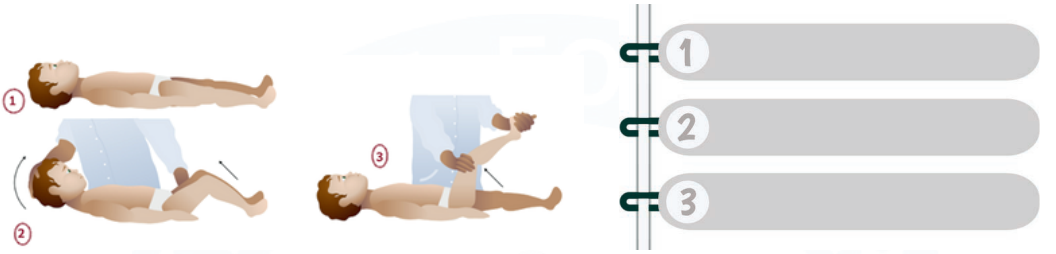
Fever with seizures: Case scenario based approach

10 month old child, 1st episode of fever + seizures (GTCS)

- Previously normal child

Why raised ICT in meningitis?

Typical features (Signs of MENINGEAL IRRITATION)



Meningitis: Etiology

Bacteria	Virus

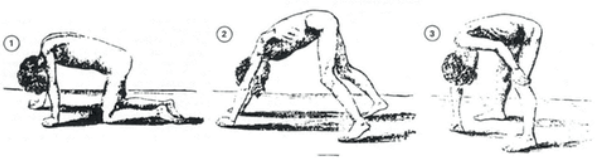
CSF analysis:

	Normal	Bacterial meningitis	Viral meningitis	TB
Appearance	Clear			
Cells				
Protein				
Sugars				

Bacterial meningitis :
Antibiotic

Which is the most common complication following bacterial meningitis?

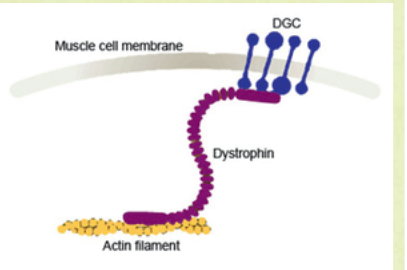
A 5 year old boy is presenting with history of delayed walking and difficulty in getting up from sitting position. Following finding is also noted and there is a family history of male relatives with similar complaints. Likely diagnosis?



Pointers to diagnosis:

Diagnosis:

Genetic defect:

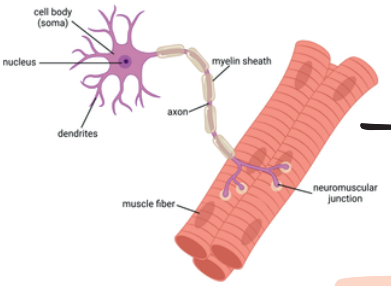


Other muscle groups affected:

- Spine muscles
- Cardiac and respiratory muscles

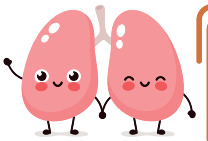
Most common cause of death?

Short Approach to a child with limb weakness



Defect anywhere in this path can result in limb weakness

child with limb weakness

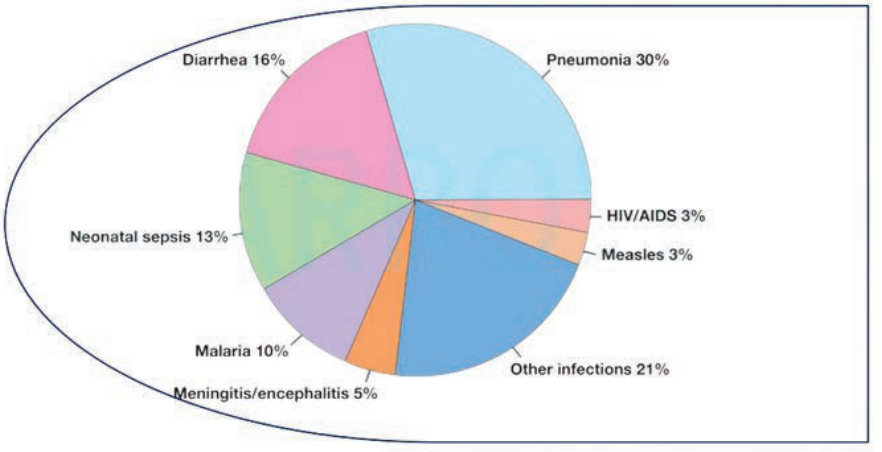


Chapter 6: Respiratory Disorders

Abnormal sounds in clinical practice

Sound	Significance
Stridor	
Wheeze (Rhonchi)	
Crepitations (Crackles)	

Under 5 Mortality in children



Pneumonia

- "Community Management ~ IMNCI"
- Hospital management

(IMNCI – Integrated management of neonatal and childhood illness)

No Pneumonia	Pneumonia	Severe Pneumonia

	Features	Management
No Pneumonia		
Pneumonia		
Severe Pneumonia		

Dangers signs	Criteria for Fast breathing (Age based)
1. _____ 2. _____ 3. _____ 4. _____	< 2 months 2 – 12 months > 12 months

Pneumonia: Etiology

- Overall m/c
- m/c virus
- In HIV infected

Pneumonia: Clinical aspects

Common features:

- fever
- breathing difficulty
- Crepitations

Radiological approach

Finding:

Organism

Treatment

Finding:

Organism

Treatment

Q) A 3 month old child presents with complaints of fever, cough, not able to feed and breathing difficulty. On examination, respiratory rate was 58/min, chest retractions were present. Management?

a)Severe pneumonia - Give antibiotics and refer immediately

b)Very severe pneumonia - refer urgently

c)Pneumonia - Start antibiotics and send child home

d)No pneumonia

1 Typical scenarios in a child with wheezing

A 6 year old child is presenting to emergency with complaints of breathing difficulty for past 1 hour. On examination, Respiratory rate is 44/min and bilateral wheeze is heard. Mother reports that child had similar episodes in past and has been using inhalers for relief.
Diagnosis:

What should be the initial management?

Essential features:

- recurrent episodes
- Lung function test:

2

A 6-month-old baby with history of URI 1 week back now presents with increasing breathing difficulty for past 2 days. There is no previous similar history. On examination, baby has low grade fever and has bilateral wheeze. Chest x ray reveals bilateral hyperinflation. What is the diagnosis?

Approach:

Typically affects children < 2 years of age and occurs during winter season

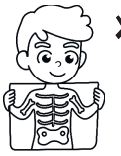
Diagnosis:

Etiology:

Features:
- fever
- breathing difficulty
- B/L wheeze

Pathophysiology:

X ray:



Management:

Role of antivirals?

---

3 A 11-month-old infant who was apparently well and playing 30 min back now presented with sudden onset of breathing difficulty. On examination, child has unilateral wheeze and dyspnea. There is no associated fever. Which condition should be suspected?

When to suspect foreign body in a child?

- Sudden respiratory complaint
- Age :

Management: Immediate removal

- Upper airway
- Lower airway

Chapter 7 - Cardiology

CONGENITAL HEART DEFECTS: Associations [Multifactorial inheritance]

Maternal associations	Genetic disorders
• Infection (TORCH) • Diabetes (overt)	• Trisomies 21 - 13,18 - • Turner syndrome • DiGeorge syndrome

Associated with defective development of 3rd and 4th pharyngeal pouches

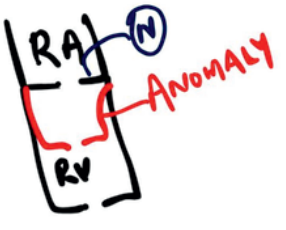
CATCH 22

- C
- A
- T
- C
- H

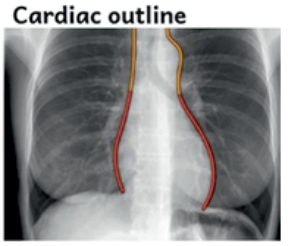
22

Heart defect associated with Lithium exposure in pregnancy?

Basic defect:



Normal appearance:

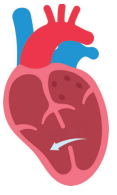


Congenital Heart defects (CHD)

- Acyanotic heart defects
  - L --> R shunts
    1.
    2.
    3.
  - Obstructive lesions →
- Cyanotic heart defects

Ventricular Septal defect (VSD):

- m/c type:



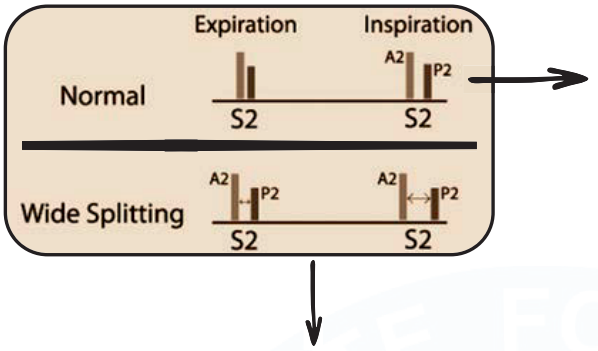
Characteristic feature:

Atrial Septal defect (ASD):

- m/c type:

Characteristic feature:

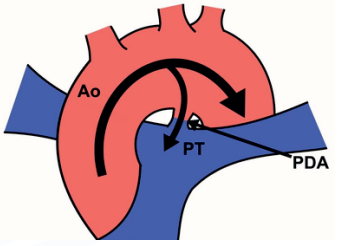
Reason for wide fixed Split of S2



Patent Ductus arteriosus (PDA):

No types

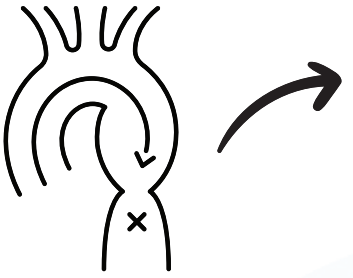
↑ incidence is noted in:



Characteristic feature:

Drug treatment in PDA

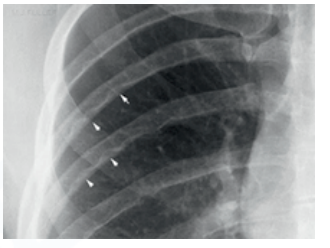
COARCTATION of Aorta:



Consequences:

Features:

Adaptation:



CYANOTIC CHD

1.
2.
3.

Others:
- Tricuspid atresia
- Truncus arteriosus

Tetralogy of Fallot (TOF)

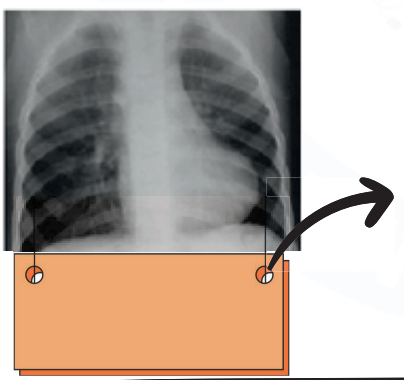
Components:

1 .....

2 .....

3 .....

4 .....

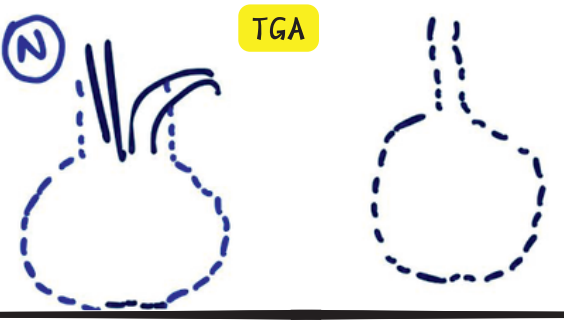


Reason?

Main determinant of cyanosis in TOF?

Finding:

Diagnosis



Finding:

Diagnosis:

Reason:

Pathophysiology

Vasculitis affecting coronary arteries:

(Most common vasculitis of childhood)

Etiology:

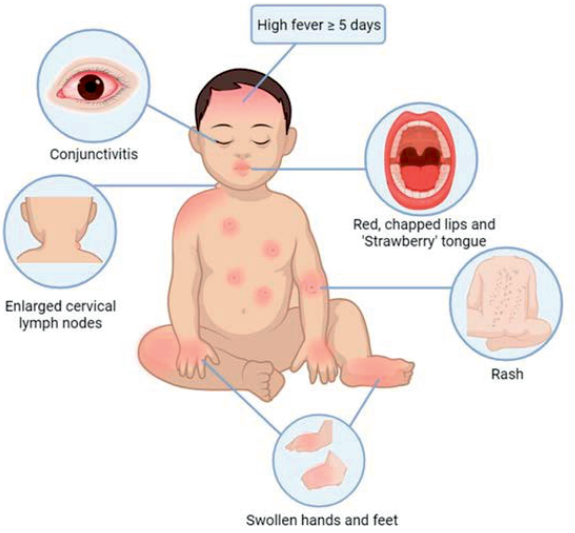
"Consequence"

Stages:		
Acute	Subacute	Convalescent
(1 - 2 weeks)	(3 - 4 weeks)	(6 - 8 weeks)

Essential feature:

Features (Acute stage):[MUCOCUTANEOUS LYMPH NODE syndrome]

A



How to remember?

C -

R -

E -

A -

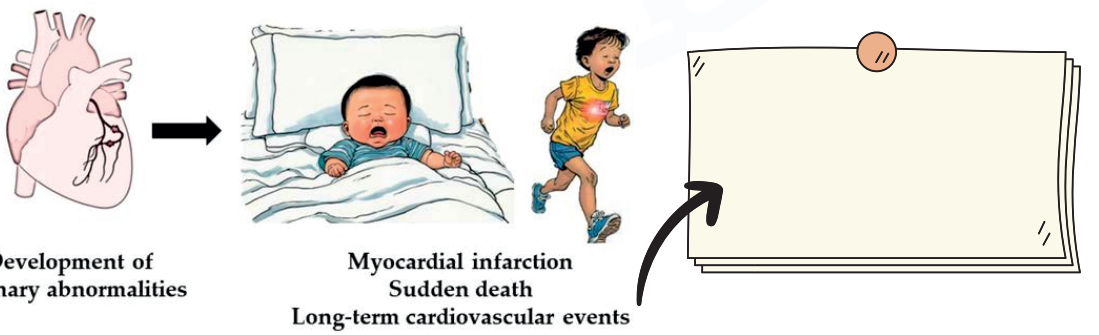
M -

Diagnostic criteria

Treatment:

Untreated Kawasaki disease

B

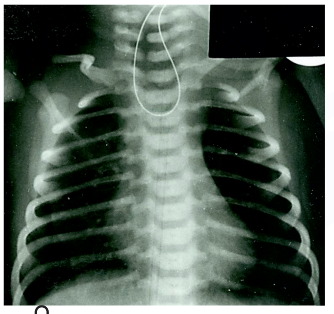


Development of coronary abnormalities

Myocardial infarction
Sudden death
Long-term cardiovascular events

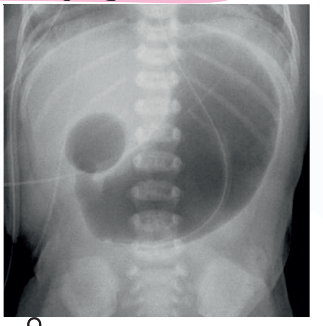
Chapter 8: Gastrointestinal disorders

Identify the condition in a neonate with repeated vomiting episodes and the x rays given below



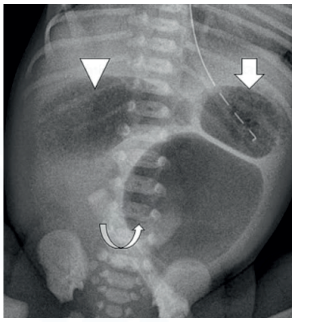
Finding

Diagnosis



Finding

Diagnosis



Finding

Diagnosis

Type of vomiting:

A 26-day old baby presented with complaints of repeated vomiting episodes for last 5 days. On examination the following finding is noted after feeds. Likely diagnosis?

Pathology

Features:

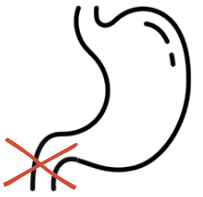
1. Vomiting
2. Mass
3.

One liners in IHPS:

Drug association:

Investigation of choice:

Consequences:



Electrolyte abnormalities	Treatment plan

Ramstedt's Pyloromyotomy

Acute Diarrhea:

- Infectious etiology

Overall m/c	m/c bacteria	Dysentery

Management of Rotavirus diarrhea:[IMNCI guidelines]

NO dehydration	SOME dehydration	SEVERE dehydration
Features: Skin pinch:	(loss of ____ mL/Kg of fluids) Features: Skin pinch:	(loss of ____ mL/Kg of fluids) Features: Skin pinch:

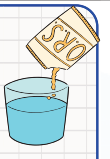
ORS ORS i.v fluids

Plan A – No dehydration	Plan B – Some dehydration	Plan C – Severe dehydration
Replace Ongoing loss	3 steps: 1) Rehydration (75 ml/Kg over 4 hours) 2) Replacing ongoing loss 3. Daily (Maintenance)fluid requirements	(Follow same 3 steps like plan B) Differences: Age: 30ml/kg (< 1yr: 1 hr; > 1 yr: 30 min) — 70 ml/kg (< 1yr: 5 hr; > 1 yr: 2 1/2 hr)

Daily (Maintenance)fluid requirements

0 – 10 Kg	
10 – 20 Kg	
> 20 Kg	

Composition



Zinc in Diarrhea

Dose

Duration

Chronic Diarrhea

Disorders:
- Celiac disease
- Cystic fibrosis

"Almost always associated with malabsorption"

Celiac disease:

- Hypersensitivity to _____
- Implicated food substances?
- "Autoantibody" damage to small intestine
- Most specific antibody:

_____

- Investigation of choice:

B -

R -

O -

W -

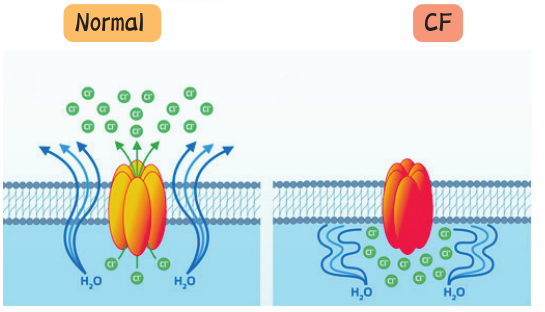
Features:
- chronic diarrhea

Treatment is lifelong gluten free diet

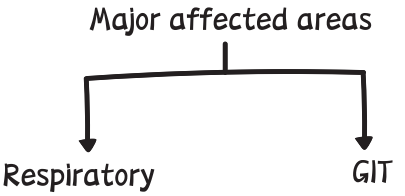
Cystic fibrosis (CF)

CFTR gene defect

Secretions in the body become thick and non functional



Cystic fibrosis (CF)



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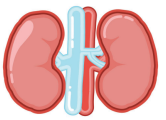
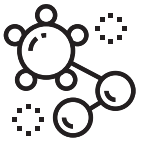


Chapter 9: Renal and Endocrine disorders



Nephrotic syndrome:

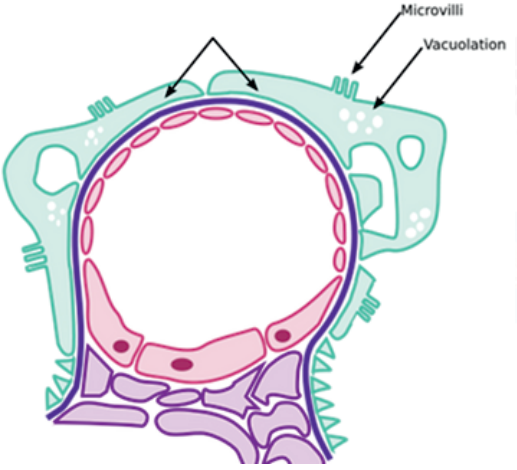
Hallmark:

- Hypoalbuminemia
- Generalised edema
- Hyperlipidemia

> How to define massive proteinuria?

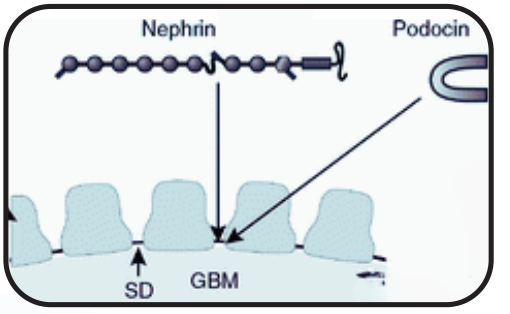
Most common Etiology:

Pathophysiology:



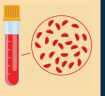
Clinical Hallmarks:

Gene defects with Nephrotic syndrome:
(Rare disorders)



Glomerulonephritis

Features of glomerular hematuria:



Other features

- Mild proteinuria
- Hypertension
- Edema

Pathophysiology:

Most common Etiology:

Lab investigations:

This condition is self limiting and occurs as SINGLE episode only

Renal vessel vasculitis:

- Idiopathic Etiology

Age group:

1
2
3

Features:

Congenital Hypothyroidism

Etiology:

Clinical features:

- Activity
- Skin
- Cry
- Bowel habits

These features are also seen in severe IODINE deficiency

Complications of untreated hypothyroidism

1 .....

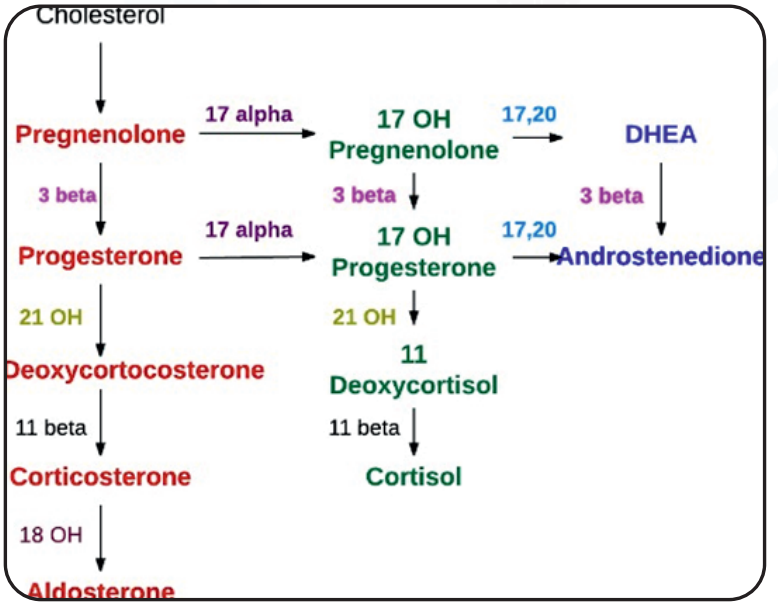
2 .....

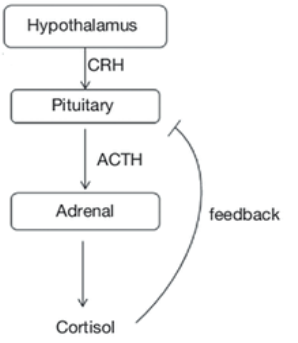
All neonates should undergo screening for congenital hypothyroidism

When to screen?

CAH (CONGENITAL ADRENAL HYPERPLASIA)

Enzyme deficiencies:





High testosterone	Low testosterone

Summary:

Common features:

- Hyperpigmentation
- Hypoglycemia

Differentiating features:

dr.singaram_med_educator

https://t.me/cropbydrsingaram

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